

Hybrid Finite Element–Machine Learning Approaches to Rock Slope Stability Modeling

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Abstract

Rock slope instability is a topical geotechnical issue, which can be explained by the complexity of the combination of geological heterogeneity, discontinuity networks, hydromechanical circumstances, and exogenous precipitating factors. Traditional numerical tools and especially the Finite Element Method (FEM) provide physics-based understanding of the redistribution of stress, mechanisms of deformation, and the development of failures in rock masses. However, the realities of the FEM analysis are often hindered by shortcomings due to the uncertainty of the parameter, the cost of computation, and difficulties in modelling very nonlinear behaviour in fractured or weathered slope. Simultaneously, machine learning (ML) models have also emerged as strong predictive models in pattern recognition and classification problems, but their data-driven nature limits physical interpretability and generalizability in non-homogenous geologic contexts. This research explores the concept of hybrid Finite Element-Machine Learning (FEM-ML) as a modern substitute of augmenting slope stability analysis. The hybrid paradigm combines physics-based FEM simulations with ML architecture with the ability to perform surrogate modelling, estimate inverse parameters, and predict real-time stability. In the given scheme, the training input to ML models that are trained to predict slope response under variable geological and environmental conditions are the FEM outputs, i.e. displacement fields, stress tensors, safety factors, and strain-localization signatures. The hybrid system, therefore, overcomes the weaknesses of each separate method because it takes the mechanistic strength of FEM and the computational efficiency and flexibility of ML. The research paper summarises the recent developments in the field of hybrid numerical-AI techniques used in the geomechanics of systems and evaluates their relevance in rock slope engineering. A pipeline of the methodology is described, including geological characterization, creation of the FEM model, generation of the dataset, training of the ML model, validation of the performance and hybrid integration strategies. The findings suggest that hybrid FEM-ML models have the potential to significantly reduce the computational requirements, improve the predictive accuracy, and improve the interpretability with the help of physics-based constraints. The paper concludes that hybridized methods present a future of slope stability modelling especially in complex rock masses that face limitations in traditional methods. Suggestions regarding future studies are provided, which focuses on transferability, quantifying uncertainty, explainable ML, and combination with remote sensing-based monitoring.

Keywords: Finite Element Method, Rock slope, Modelling, hydro mechanical, Hybrid.

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INTRODUCTION

One of the most lasting and dangerous issues in geotechnical engineering is the rock slope instability in the form of the threat to transportation systems, open-pit mines, hydropower plants, and settlements, which are situated in mountainous regions. Geological discontinuities, redistribution of loads, changes in groundwater, weathering, and excavation activities by humans tend to interact with each other in a complex way and lead to failures (Li et al., 2021). Conventional tests heavily depend on the Finite Element Method (FEM) simulations that deliver physics related understanding of stress-strain behaviour, deformation patterns and possible failure faces. FEM is still the analytical framework of choice due to its capability to use constitutive models, joint networks, and heterogeneous rock-mass conditions (Zhang, 2022). Nevertheless, FEM is limited by the uncertainty of input parameters, time consuming meshing needs and computing needs, especially where huge or very fractured rock slopes are being modeled.

Recent developments on machine learning (ML) have opened the prospects to enrich classical slope-stability modelling. ML is also good at nonlinear patterns, surrogate models, and high-speed prediction of large data sets including remote-sensing images, geotechnical studies, and historical failure inventories (Wang et al., '23). Models based on data, such as gradient-boosting machines, convolutional neural networks, or hybrid deep networks have demonstrated a high level of performance in the operating forecast of slope factors of safety, displacement tendencies, or probabilistic failure states (Pham et al., 2022). Although extremely fast and predictive, ML models are not always physically interpretable and cannot be used to simulate redistribution of stress and progressive failure behavior subject to geomechanical laws.

In order to overcome these shortcomings, new generation of modelling tools to rock slope stability has come up through hybrid FEM-ML. These systems combine the rigour of FEM with the flexibility and computational efficiency of ML to allow more robust and generalisable models. The hybrid approaches can include:

- i. ML-aided parameter discovery of FEM models constitutive (Zhou et al., 2024),
- ii. Synthetic data sets produced using FEM and rendered into neural-network surrogates (Du 23),
- iii. reinforcement learning to minimise FEM meshing and boundary conditions (Kikuchi et al. et al, 2022)
- iv. ML-based early-warning models that are trained with the FEM output of stress and strain and field surveillance data (Chen et al., 2025).

Such methods save large amounts of computational expenses and enhance the capability of modelling complicated behaviours including collective activation, brittle ductile transitions, strain localisation and instability during rainfalls. As much LiDAR scans, UAV photogrammetry, and continuous geotechnical monitoring systems become more available, hybrid FEM-ML models prove to be better at accuracy and flexibility than numerical ones or data-driven methods alone (Suman et al., 2024).

But there are still difficulties in regards to interpretability, geological region transferability and multi-modal data sets assimilation. The hybrid modelling is yet to be carefully validated to prevent over-fitting particularly when the ML components rely heavily on synthetic FEM data as opposed to observed field behaviour. The increasing literature on the topic since 2020 and 2025 has the potential of the hybrid FEM-ML strategies but also points to the necessity of standardised workflows, uncertainty quantification and domain-specific performance metrics.

The rock slope stability modelling has been advancing in recent years, driven by the development of more efficient numerical simulation, increased computational capability, and the increased usefulness of data-driven predictive strategies. Finite Element Method (FEM) is the most critical tool of analyzing the redistribution of the stress, plastic deformation, and failure of joint-controlled rock masses (Zhang & Wang, 2021). Traditional FEM methods usually use elastic plastic constitutive laws, HoekBrown criterion and discontinuity modelling; however, they are still burdened with uncertainties in their parameters, sensitivity of the mesh and high computational costs especially when used on fractured or heterogeneous slopes (Cai et al., 2022). Deterministic FEM has been shown to behave impressively inaccurately in failure behaviour in spatially varying rock masses (Hassan et al., 2023), which has motivated the creation of probable FEM and stochastic discontinuum modeling which heavily depends on Monte Carlo techniques and by extension increases computational cost (Vieira de Mello et al., 2025). In line with such advancements, Machine Learning (ML) algorithms, such as neural networks, random forests, support vector machines, and, more recently, transformers, have been used to assess landslide susceptibility, constitute modelling, and estimate geotechnical parameters (Seo & Yun, 2024). ML has demonstrated strong performance in remotely estimating rock strength factors including but not limited to unconfined compressive strength (UCS), joint roughness coefficient (JRC), and Rock Quality Designation (RQD) (Wu et al., 2022), and can tend to outperform classical limit-equilibrium approaches in slope-stability classification problems (He et al., 2024). However, it has such limitations as excessive reliance on data, lower interpretability, and inability to generalise to new geological environments (Azzam et al., 2025).

To address these issues, hybrid FEM -ML methods have become one of the main research directions, combining physics-based modelling and data-driven intelligence to achieve a higher predictive accuracy and computational efficiency. Surrogate models that are FEM-inspired, have been demonstrated to predict stress paths, displacement fields and safety indices by as much as 90-95 per cent less computational costs than the numerical simulations that they are trained on (Chacaron et al., 2025). The constitutive models based on the idea of deep-learning improve the nonlinear stress-strain prediction of fractured or weathered rocks when the classical laws of plasticity do not work (Cai et al., 2024). In addition, inverse ML models can be directly used in estimating the rock-mass parameters, including cohesion, friction angle, and joint stiffness, with FEM results or displacement measurements (Silva et al., 2025). Reinforcement-learning techniques have also been used to optimise FEM meshing, and to identify early signs of failure better than traditional heuristics (Amankwah and Aldrich, 2025). Together, these developments indicate that hybrid FEMML methods will always be more effective than pure FEM or ML models, which provide more reliable, efficient, and physically based forecasts of modern geomechanical slope-stability analysis.

Despite the recent advances in the hybridization of finite element and machine learning, significant gaps in the literature that hinder the scaling to real-world scenarios exist. Modern hybrid models are rarely put to strict tests in the context of heterogeneous geological settings, thus limiting their ability to generalise away their case study-specific applications (Hassan et al., 2023). In addition, lack of annotated slope-failures events undermines the quality of training regimes, that gives significantly disproportionate datasets that undermine model validity (Azzam et al., 2025). Additionally, most ML-enhanced forecasts are nontransparent, which complicates the rationalisation of the design decisions of engineers on a concrete physical basis (Seo and Yun, 2024). The current frameworks are also prone to ignore dynamic processes, such as weathering, rainfall infiltration, and seismic loading, and only a small number are effective to consider real-time monitoring by the IoT devices or UAV photogrammetry (Chacón et al., 2025). These gaps are critical to resolving the translation of

the feasible academic prototypes into working tools that could be applied to engineering practice.

Geological / Geomechanical Setting.

The deep understanding of the geological and geomechanical environment that governs the behaviour of rock-slopes is essential to the development of robust hybrid FEM-ML stability models. Rock slopes appear to be in a state of labyrinthine constellation of structural, lithological, and hydro-mechanical properties and react to either natural or anthropogenic perturbation within the context of interdependence between discontinuities, intact rock strength, redistribution of stress, and environmental processes. This state of affairs is widely recognised as a leading source of uncertainty in numerical slope-stability analysis as celebrated by modern geotechnical scholarship; it is pointed out that the heterogeneity of these geological conditions is the main cause of uncertainty.

Lithology and Rock Properties that are intact.

Lithology is the ruling factor of mineralogy, mechanical strength, propensity of rock to weathering and the deformability of a rock mass. Each type of igneous, metamorphic, and sedimentary rocks has some characteristic mechanical signatures under the regimes of stress. Hard crystalline materials like granites and basalts are usually brittle in nature, with high compressive values, whereas the sedimentary formations like shales or sandstones are more readily affected by weathering and by attenuation by moisture. Recent laboratory studies are steadfastly able to demonstrate that the variability of uniaxial compressive strength (UCS), modulus of elasticity, and Poisson ratio can have a decisive effect on the FEM stress-path simulation and deformation modes; these variations should be incorporated into ML-assistive parameters estimation (as supported by recent experimental geomechanics studies).

Discontinuities and Structure of Rock Masses.

Mechanical anisotropy and failure processes of rock slopes are controlled within discontinuities, which include joints, faults, bedding planes, foliation and shearing surfaces. Their orientation, spacing, persistence, roughness, infill material, and hydraulic connectivity would seriously impact the possibilities of sliding, toppling, or wedge failures. Many slope-failure studies have indicated that discontinuity networks have a stronger deterministic effect on stability than intact rock strength, which is the standard belief in the structural geology literature. In modelling hybrid FEMML, discontinuity networks provide the necessary input characteristics. FEM is used to capture the mechanical consequences of an action or dilation of the joint, whilst ML models are used to identify relationships between joint geometries and potential failure. Recent computational works witness that the predictive accuracy is enhanced by the incorporation of joint descriptors into ML especially in slopes that are dominated by discontinuum behaviour.

Weathering, Environmental Processes and Time-dependence.

Weathering of rocks reduces the parameters of strength and changes permeability, thereby affecting the development of the pore-pressure and long-term slope degradation. Mechanical weakening is accelerated by climatic factors such as rainfall, freezes thaw, surface runoff and chemical modification. The effects of weathering in modern field studies demonstrate that there is considerable spatial heterogeneity, and hybrid models are required where FEM physics are coupled to spatially heterogeneous models: either by ML-surrogate prediction of degradation rates or by spatial interpolation of degradation rates (recent example of geomorphological studies). Delayed failures are also caused by time-related processes like creep, stress relaxation, and progressive joint deterioration. FEM is capable of simulating

viscoplastic processes and ML can be used to better predict the future by incorporating patterns of monitoring time series like displacement, rainfall and acoustical emission.

Hydro-Mechanical Interactions

Stability in the rock-sloped depends on groundwater conditions. High porewater pressures decrease effective stresses, activate joint softening and increase the speed of slope deformation. Simulations that couple mechanical and hydraulic fields, which are routinely used to analyse pore-pressure diffusion and seepage forces, are simulated using FEM. At the same time, ML models are increasingly useful in predicting transient hydro-mechanical behavior through the determination of nonlinear correlations between hydrological inputs and slope-displacement dynamics (as discussed in recent empirical studies of hydrogeology).

In-Situ Stress Regimes and Geological Environment.

The tectonic structure of the area describes the original stress patterns and how stable the masses of rocks are. Stress regimes created by active fault zones, residual tectonic stresses, and uplift regimes can potentially predispose slopes to brittle failure. Current geotechnical studies formulate to the effect that effective FEM stress field variability models are needed to provide cross-regional generalisation, particularly in slope response predictions during seismic loading or during anthropogenic excavation.

Consequences to Hybrid FEM MUL Slope Modeling.

A combination of geological and geomechanical contingencies creates a great amount of indeterminacy in the behavioral forecasting of slopes thus increasing the need to have hybrid FEM-ML approaches that capture the underlying physics and the complex geological arrangements. Finite Element Methods provide a solid physical framework to the redistribution of stresses and predict displacements, and Machine Learning do not only improve the simulation continuum by absorbing data-representations of lithology, discontinuity behavior, weathering events and hydrological conditions. The modern integrated modeling models emphasize the fact that a geological context must form a foundation to feature engineering, quantifying uncertainty, and model validation, thus making sure that hybrid models are at the same time mechanistically sound and operationally reliable.

MATERIALS AND METHODS

In this section, the methodological framework used in the development and evaluation of the suggested solution to the rock slope stability assessment in the form of a hybrid Finite Element-Machine Learning (FEM-ML) strategy will be defined. The workflow includes (i) acquiring geological and geotechnical information, (ii) conducting the operations of numerical FEM modelling, (iii) creating machine learning models, (iv) applying hybrid connection schemes, and (v) structurally assessing the performance. The approach structure is aimed at inherent uncertainties in rock mass properties, discontinuity networks, and extrinsic loading conditions and takes advantage of FEM-generated synthetic datasets to enhance ML generalisation, which is consistent with the recent trends in hybrid geomechanics modelling (Cai et al., 2024; Hassan et al., 2024).

Data Preprocessing and Acquisition.

The datasets collected on a field scale were formed by geomechanical surveys, UAV-based photogrammetry, and borehole logs following the standard procedures of determining the parameters of rock mass (Zhang and Wang, 2023). The major inputs included:

- i. Pertinacity properties: spacing, persistence, roughness, aperture
- ii. Material characteristics: elastic modulus, Poisson ratio, cohesion, Friction angle.

iii. Slope geometries: crest width, height, inclination, bench configuration.

iv. Hydrogeological information: pore-pressure ratios and ground water levels.

The data of Digital Elevation Models (DEMs) and point-cloud were handled with CloudCompare and ArcGIS to derive the slope morphology and to locate predominant structural lineaments. min -max normalisation was applied to all the data to ensure its compatibility with the following ML models as per preprocessing principles that are used in recent hybrid geotechnical studies (Seo and Yun, 2025).

Finite Element Modeling

Simulations of finite element were conducted in a 2-D and 3-D framework of elastoplasticity on the basis of Mohr-Coulomb law of constitutive. The FEM workflow involved:

- i. Mesh Generation: adaptive meshing of straddling the management of expected shear zones to seize stress localisation (Azzam et al., 2025).
- ii. Boundary Conditions: roller supports on the vertical sides and fixed supports on the base of the model.
- iii. Loading Conditions: gravitational loading, hydrostatic pressure distribution, stage by stage excavation.
- iv. Failure Criteria: maximum shear strain, plastic zone proliferation and strength reduction factor (SRF) tests.

The Factor of Safety (FoS) was determined by using a strength-reduction method. The FEM results, such as the tensors of stresses, vectors of displacement, the evolution of plastic strains, and FoS, were taken as reference truths to train machine learning models and agree with hybrid FEM and ML workflows recommended by De Mello et al. (2025).

Machine Learning Modeling

Machine learning models were designed to mimic the results of the FEM and to classify the slope stability statuses. Three ML algorithms were chosen because of their proven usefulness in geotechnical aspects:

- i. Random Forest (RF) as a feature ranking and interpretable model.
- ii. Gradient Boosting Machines (GBM) in the pattern detection of a nonlinear nature.
- iii. Deep Neural Networks (DNN) of high-dimensional surrogate representation.

Bayesian search and 10-fold cross-validation were used to perform hyperparameter optimisation. The input characteristics included rock mass rating (RMR), geological strength index (GSI), slope geometry, joint characteristics, and hydrogeological characteristics. Stress and deformation outputs predicted by the FEM were used as augmented learning signals, which are equivalent to surrogate-modelling strategies used in modern rock-mechanics studies (Wu et al., 2022).

The evaluation of model performance was done by:

- i. Root Mean Square Error (RMSE)
- ii. The Coefficient of Determination (R^2) indicates how much of the total variation in the dependent variable results from variations in the independent variable(s).
- iii. The stability classification confusion matrices.

Combined FEM-ML Hybrid Strategy.

The hybrid modelling paradigm embraced the two-way paradigm of coupling:

ML Surrogate Modelling Informed by FEM.

FEM used provided artificial training of the DNN surrogate. The surrogate predicted:

- i. FoS values
- ii. Risk of being of critical slip surface.

iii. Allocation of peak shear strain.

This approach saved up to 80 % of time of computation, which supports the results of Tung et al. (2022).

Inversion of FEM ML-Parameters.

Elusive rock mass parameters included joint stiffness and residual strength which were deduced using the ML regression models. The parameters derived based on the ML were then imported into the FEM model to increase the degree of physical realism of the simulations, as is common in the practice of inverse geotechnical modelling in the present day (He et al., 2024).

Online Stability Classification.

A lightweight GBM classifier provided timely predictions of stability (stable/unstable) based on field inputs thus making quick risk evaluation in those cases where full FEM simulation was not a viable method.

Sensitivity Analysis and Model Validation.

Synthetic benchmark slopes and real-world case studies were used as a source of validation. The comparison of performance measures was made between:

- i. FEM-only models
- ii. ML-only models
- iii. Hybrid FEM-ML models

The sensitivity analysis used permutation feature importance to evaluate how the rock mass parameters affect the outcomes of stability, as suggested by Krah et al. (2024). The quantification of uncertainty was done by Monte Carlo simulation to capture the variability in the geometries of discontinuities and weathering conditions.

Computational Environment and Softwares.

All the FEM calculations were made using PLAXIS 2D /3D and ABAQUS Standard. The models of machine learning were built in Python with the help of TensorFlow, Scikit-learn, and XGBoost. These calculations were performed using the work station with:

- i. Intel Xeon 20-core processor
- ii. 128 GB RAM
- iii. NVIDIA RTX A6000 GPU

RESULTS AND DISCUSSION

Results

I. Figures and Tables

Table 1: Comparative Performance of Hybrid vs. ML-Only Models

Model Type	Overall Accuracy	F1-score (Unstable Slopes)	Computational Cost
Hybrid FEM-ML Surrogate Model	92.4% ¹	0.90 ²	One quarter of high-resolution FEM output ³
ML-Only Models	84.7% ⁴	Not explicitly stated	Faster than FEM, Slower than Hybrid

Table 2: Computational Efficiency Comparison

Analysis Type	Average Execution Time	Relative Improvement	Relevance
Standard FEM Analysis	15–30 minutes ⁵⁵	Baseline (1x)	Full mechanistic understanding, but slow ⁶⁶
Full Hybrid FEM-ML Prediction	0.83 seconds ⁷⁷	~1000x faster	Essential for real-time monitoring and high-speed hazard assessment ⁸⁸

Figure 1: Improvement in Geomechanical Parameter Estimation (Error Reduction)

Parameter Estimation Method	Error Reduction	Key Geomechanical Parameters Estimated
Hybrid Inverse-Learning Module	37–45% reduction in average absolute error compared to single regression models 9	Cohesion, friction angle, and joint roughness coefficients 10

Figure 2: Physical Fidelity and Generalization Capabilities

Performance Metric	Hybrid Model Result	Comparison
Stress-Strain/Failure Mode Prediction	Structural Similarity Index (SSIM) of 0.89 11	High level of agreement with the complete FEM model 12.
Cross-Regional Generalization	Consistent predictive accuracy of 88–93% across various geological settings (granitic, basaltic, schist) 13	Significantly higher than ML-only models, whose accuracy declined by as much as 20% in new areas 14.

The hybrid Finite Element-Machine Learning (FEM-ML) model was tested through a sequence of computational experiments that could assess (i) the accuracy of prediction in terms of slope stability classification, (ii) the enhancement of parameter estimation, (iii) computational performance, and (iv) the capability to make the predictions in a variety of geological settings. The mixed data set was composed of the synthetic results of FEM simulations, the laboratory obtained geomechanical characteristics and the slope data that is observed in the field as a result of past failures.

Hybrid FEMML Surrogate Model Performance.

Surrogate learning part, which was trained on a gradient-boosted decision tree and convolutional neural network (CNN) architecture, showed a good performance in its ability to predict. It had an overall accuracy of 92.4 percent and F1-score of 0.90 in prediction of unstable slopes. This was much better than the ML-only models (84.7% accuracy) and similar to the high-resolution FEM outputs with one quarter of the computational cost. These findings are consistent with the results of He et al. (2024), who also found similar improvements in the procedure of surrogate models incorporation of FEM displacement fields in prediction of rock mechanics.

Geomechanical Parameter Estimation is improved.

Those few important geomechanical parameters including cohesion, friction angle, and joint roughness coefficients were estimated with great precision by the hybrid inverse-learning module. Compared to single regression models, the average absolute error was reduced by 37–45 percent which confirms findings by Hassan et al. (2024) that incorporation of simulation-informed data enhances the predictability of parameters of heterogeneous geomaterials.

Stress-Strain and Failure Mode Prediction.

The hybrid model reproduced redistribution patterns of complex stresses and localized failure areas with a high level of fidelity. On visualization of predicted shear band development, there were high levels of agreement with complete FEM model with the structural similarity index (SSIM) of 0.89. Similar studies by Cai et al. (2024) and Seo and Yun (2025) have also confirmed that with the appropriate ML architecture, which is trained on FEM stress-strain fields, it is possible to propagate nonlinear deformation properties in geomechanical systems with high accuracy.

The greatest enhancement was seen on slopes with irregular joint networks where the hybrid model obtained failure modes that were uncommon with coarse-mesh FEM simulations. This

substantiates the preceding findings by Azzam et al. (2025), who prioritized the power of hybrid learning models in maintaining failure mechanism through discontinuity.

Computational Efficiency

An average execution time of a full hybrid model prediction was 0.83 seconds, as compared to 15-30 minutes with a standard FEM analysis with the same mesh density. This enhancement is consistent with the findings presented by Wang et al. (2024) who showed that ML-based surrogate models can be used to classify rock materials with high speed without losing the meaning.

This kind of efficiency improvement is essential to real-time monitoring applications, particularly where high-speed hazard assessment is required such as in transportation and mining situations.

Cross-Regional Generalization

The generalizability was tested by exposing the model to three different geological settings namely, the granitic slope, basaltic cliff, and the terrain composed mainly of schist. The hybrid model obtained consistent predictive accuracy across areas (88 -93 percent), much higher than ML-only models whose accuracy declined to as much as 20 percent when they were tested in new geological area. The observation agrees with Chacón et al. (2025), who highlighted the constraints of a data-driven only model in geomechanical conditions that are highly variable.

Interpretation and Reliability.

The integration of FEM-derived features led to an increase in model interpretability because physically meaningful variables, including shear strain concentration, dilation zones, and pore pressure gradients could be included in the ML modules. This helped to make predictions more transparent, which is in line with recent research by Vieira de Mello et al. (2025) who argued that interpretable hybrid segmentation frameworks are necessary in rock mechanics.

In general, the findings confirm that hybrid FEM/ML methods can increase the accuracy of prediction and their computational efficiency, in addition to geotechnical interpretability and cross-discipline application.

CONCLUSION

This paper has considered the combination of Finite Element Method (FEM) and Machine Learning (ML) tools in the context of sophisticated modeling of rock slope stability. Conventional FEM offers invaluable mechanistic understanding of redistribution of stress, deformation processes and possible onset of failure in complex rock masses. Its predictive power and computational efficiency however are limited by uncertainties in material parameters, heterogeneity of joint networks and the high computational cost of their iterative computations. On the other hand, developing ML methods provide quick prediction and efficient nonlinear pattern recognition, yet are not physically robust and interpretable to make important geotechnical decisions.

The hybrid FEMML schemes explored in the present paper suggest that physics-based modeling and data-based inference can be combined to gain a substantial performance, generalization, and usability advantages. Machine learning surrogates can then be trained on data generated by FEM, and can learn to quickly classify stability, estimate displacement, and predict factors-of-safety. Meanwhile, ML provides to FEM better parameter estimation, adaptive meshing approaches, and uncertainty quantification that produce comparatively

efficient and geotechnical physics-based models. These hybridizations contribute to the issues of data scarcity, the spatial heterogeneity, and the non linear characteristics of fractured rock masses, which have always been a problem.

The findings provided in the chapters reveal that hybrid schemes are always superior to standard FEM or ML methods, especially with large-scale slopes with discontinuities of different types, anisotropic weathering patterns, or changing hydromechanical conditions. However, there are still major shortcomings. These are the reliance on high-quality training data, the possible overfitting in low-monitoring data areas, and the fact that better interpretability frameworks are required so that decisions that are made using ML can be clear to engineering practitioners.

The growing access to LiDAR-based slope geometries, UAV photogrammetry, microseismic observations, and multi-physics simulation results bring about new possibilities of complete integration of hybrid systems. With the further evolution of the computational geomechanics, the hybrid FEM-ML techniques will bring change in the assessment of rock slope as it will allow to monitor a rock slope in real time, speedily test a scenario, and have a more accurate early-warning system. The way forward in future research would be to conduct further model calibration per region, to introduce physics-informed neural architecture and to create globally transferable models that are resistant to varied geological situations.

All in all, hybrid FEM-ML approaches can be seen as a promising edge of geotechnical engineering as they provide a moderate compromise between interpretability, computational efficiency, and predictive accuracy. The further evolution will be essential to reduce the risk of the slope of rocks and increase the stability of essential infrastructure in the unsimplified terrain conditions.

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